LESSON PLAN · SCRATCH (GR 6–8) ADVANCED: ALGORITHMS LAB

Lesson 12: Capstone: Algorithm App

Module 4 | 90 minutes | Capstone project | Grades 6–8 · ages 12–14

Objective: Design, build, test and present an app built around an algorithm (searching, sorting, ciphers or recursion), with the matching Python written alongside.

Success criteria

- My app is built around an algorithm in a procedure.
- I measured its efficiency.
- I wrote the Python version and presented it.

Key vocabulary

- **pseudocode**: An algorithm written in plain, structured steps.
- **efficiency**: How the work grows with the input.
- **implementation**: Turning an algorithm into working code.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)
- Printed project brief

Timed lesson plan (90 min)

Time	Stage	What happens
10 min	**Session 1 · Pseudocode**	Choose an algorithm; teacher checks the pseudocode.
30 min	**Session 1 · Core**	Algorithm in a procedure, tested with known inputs.
5 min	**Session 1 · Save**	Save; plan the app around it.
20 min	**Session 2 · App and Python**	Input, output, validation; write the Python.
10 min	**Session 2 · Measure & test**	Step counts for several inputs.
15 min	**Session 2 · Submit & present**	Reflection, submit, 1-minute demos.

Unplugged warm-up

Pseudocode pairs (5 min): write pseudocode; a partner “executes” it with cards.

Teaching the idea

Assesses algorithmic thinking (search, sort, strings/ciphers, recursion), modular design, efficiency measurement, and translation to Python. Pseudocode → blocks → Python.

Analogy: A chef presenting a signature dish and its recipe.

Live demo: Show pseudocode, blocks and Python for bubble sort side by side.

Questions to ask

Ask	Listen for
What algorithm did you choose and why?	Learner-specific.
How did you measure efficiency?	Step counts.
How does your Python differ from the blocks?	Syntax, indexing.

Differentiation

Support
Extend the leaderboard or cipher project with input, validation and step counts.

Stretch
Compare two algorithms in one app (e.g. bubble vs a smarter sort).

Common misconceptions

Misconception	How to fix it
A capstone needs a brand-new algorithm	Applying a known algorithm well is the goal.
Python must be perfect	Clear, mostly-correct Python shows understanding.

Assessment

Project rubric (also on the project brief):

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Algorithm	Incorrect	Works for some inputs	Correct for all tested inputs	Correct and efficient, edge cases handled
Design	One script	Partly modular	Algorithm in a procedure	Clean, well-named procedures
Efficiency	Not measured	One count	Counts for several inputs	Explains growth (e.g. n^2)

Python	None	Fragments	Mostly correct translation	Accurate, idiomatic Python
Presentation	None	Demo only	Demo and explanation	Clear walkthrough of algorithm, blocks and Python

Exit ticket
One thing your step counts showed.

Homework
Explain your algorithm to someone at home without using the computer.

Answer key — Kitabash activities

1. Which algorithm would you use? (perform · categorize)

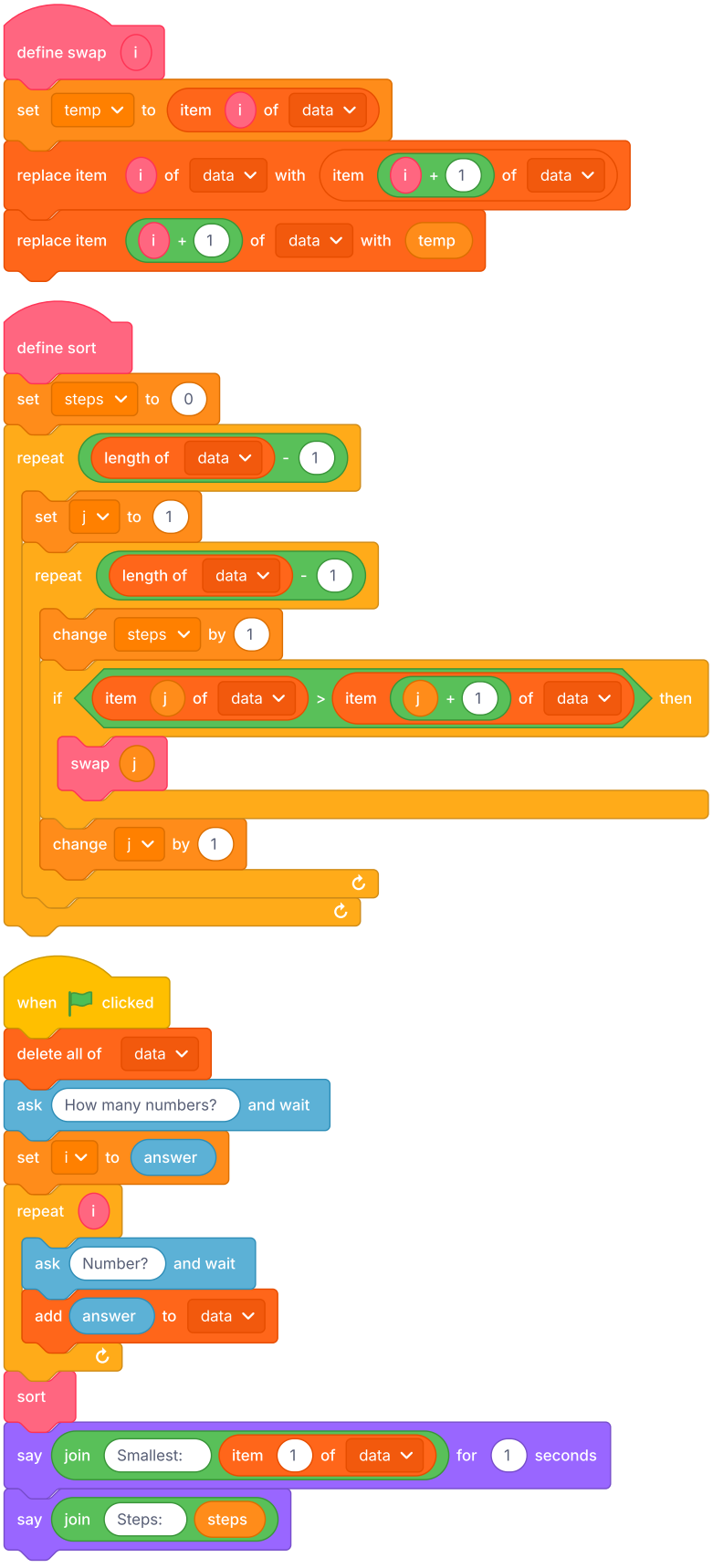
Binary search: Find a name fast in a sorted contact list
Bubble sort: Rank players by score, Put words in alphabetical order
Caesar cipher: Hide a message from classmates
Recursion: Draw a snowflake made of smaller snowflakes

2. Capstone: Algorithm App (probe · code lab)

Goals checked: Algorithm in a procedure · Loops within loops · Takes input · Reports a result

Example project:

Coder



Answer key — worksheet

1. **Key words:** pseudocode = An algorithm written in plain, structured steps. · efficiency = How the work grows with the input. · implementation = Turning an algorithm into working code.
2. **Which algorithm would you use?:** Binary search: Find a name fast in a sorted contact list · Bubble sort: Rank players by score, Put words in alphabetical order · Caesar cipher: Hide a message from classmates · Recursion: Draw a snowflake made of smaller snowflakes
3. **My plan:** A clear plan that covers every requirement.
4. **Checklist:** All requirements met.
5. **Two stars and a wish:** Kind, specific feedback and an improvement.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.